

OLS Estimator via Singular Value Decomposition

This solution originated as a multiple choice problem from Linear Algebra preparatory work for the MS in Data Science & AI program at USF, beginning summer 2026. What began as selecting the correct answer evolved into a more rigorous proof and learning exercise, with careful attention to the underlying linear algebraic properties and common points of confusion that a terse multiple choice format does not surface.

Problem

Given the OLS regression model $Y = X\beta + \epsilon$, where $X \in \mathbb{R}^{n \times p}$ with $n > p$, $\beta \in \mathbb{R}^{p \times 1}$, and ϵ is mean-zero noise, the OLS estimator is

$$\hat{\beta} = (X^T X)^{-1} X^T Y.$$

Express $\hat{\beta}$ in terms of the SVD of X , where $X = UDV^T$.

Background

Key Result (Orthonormal Vectors). *A set of vectors $\{v_1, \dots, v_k\}$ is **orthonormal** if*

$$v_i^T v_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

That is, every vector has unit length, and distinct vectors are mutually perpendicular. The inner product $v_i^T v_j$ measures the projection of v_i onto v_j ; orthonormality says each vector has no component in any other direction in the set. This is the natural generalization of the standard basis vectors $e_1, \dots, e_k$.

Key Result (Orthonormal Matrix — Square Case). *A square matrix $M \in \mathbb{R}^{k \times k}$ is **orthonormal** (orthogonal) if its columns form an orthonormal set. Writing out the column inner products in matrix form, this is equivalent to*

$$M^T M = M M^T = I_k.$$

This has a powerful consequence for inverses: comparing with the definition $MM^{-1} = I_k$, we immediately read off that $M^{-1} = M^T$. In other words, for an orthonormal square matrix, the inverse is free to compute — it is simply the transpose. A further consequence used below is that $(M^T)^{-1} = (M^{-1})^T = (M^T)^T = M$, i.e. the inverse of the transpose is the original matrix.

Key Result (Orthonormal Matrix — Tall Case). *For a tall matrix $M \in \mathbb{R}^{n \times p}$ with $n > p$ and orthonormal columns, the column orthonormality condition gives only the weaker one-sided identity*

$$M^T M = I_p.$$

The product MM^T is $n \times n$ but has rank at most $p < n$, so $MM^T = I_n$ is impossible — I_n has full rank n , while MM^T can never be. Geometrically, MM^T acts as a map from $\mathbb{R}^n$ to $\mathbb{R}^n$, but its image is only the p -dimensional column space of M — a proper subspace of $\mathbb{R}^n$. Any vector v orthogonal to the column space of M satisfies $M^T v = 0$ by definition of orthogonality, and therefore $MM^T v = M(M^T v) = M \cdot 0 = 0$, so MM^T sends all such vectors to zero and cannot act as the identity on all of $\mathbb{R}^n$. The one-sided identity $M^T M = I_p$ is nonetheless sufficient for our purposes: it allows us to cancel $M^T M$ to I_p whenever this product appears, which is the key cancellation exploited in Steps 1 and 3 below.

Key Result (Diagonal Matrices and Their Inverses). A diagonal matrix $D = \text{diag}(\delta_1, \dots, \delta_p)$ with all positive diagonal entries is invertible, with

$$D^{-1} = \text{diag}(\delta_1^{-1}, \dots, \delta_p^{-1}).$$

This follows directly from the definition: $DD^{-1} = \text{diag}(\delta_1 \cdot \delta_1^{-1}, \dots) = I_p$. A useful consequence is that for any integer power k , $D^k = \text{diag}(\delta_1^k, \dots, \delta_p^k)$, so in particular $(D^2)^{-1} = D^{-2}$ and $(D^{-2})D = D^{-1}$, mirroring the familiar rules of scalar exponents. Note also that diagonal matrices are symmetric ($D^T = D$) and commute with one another.

Key Result (Inverse of a Product). For square invertible matrices A, B :

$$(AB)^{-1} = B^{-1}A^{-1}.$$

This reversal of order is necessary: AB first applies B then A , so to undo it one must first undo A (apply A^{-1}) and then undo B (apply B^{-1}). The rule extends to three factors as $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$, which is applied in Step 2 below. Crucially, this rule requires all matrices involved to be square and invertible — it cannot be applied to X directly since X is not square.

SVD Components

In the thin SVD $X = UDV^T$:

- $U \in \mathbb{R}^{n \times p}$ has orthonormal columns, so $U^T U = I_p$ (tall case above).
- $D \in \mathbb{R}^{p \times p}$ is diagonal with positive singular values $\delta_1, \dots, \delta_p$, hence invertible and $D^T = D$.
- $V \in \mathbb{R}^{p \times p}$ is square and orthonormal: $V^T V = V V^T = I_p$, so $V^{-1} = V^T$ and $(V^T)^{-1} = V$.

Derivation

Step 1: Compute $X^T X$ using the SVD

Substituting $X = UDV^T$ and using $D^T = D$ (diagonal matrices are symmetric):

$$X^T X = (UDV^T)^T (UDV^T) = VD^T U^T \cdot UDV^T = VD \underbrace{(U^T U)}_{=I_p} DV^T = VD^2 V^T.$$

The cancellation $U^T U = I_p$ is the tall-case orthonormality of U : the columns of U are orthonormal, so their pairwise inner products collapse to the identity.

Remark. A natural first instinct is to attempt to simplify $(X^T X)^{-1} X^T$ by writing $(X^T X)^{-1} = X^{-1} (X^T)^{-1}$, and then cancelling $(X^T)^{-1}$ against X^T to obtain X^{-1} . However, this reasoning is invalid because X^{-1} does not exist for a non-square matrix. The inverse of a matrix A is defined as the matrix B satisfying $BA = AB = I$ with the same identity matrix on both sides. For a tall matrix $X \in \mathbb{R}^{n \times p}$ with $n > p$, this is impossible: XB is $n \times n$ and has rank at most $p < n$, so $XB = I_n$ can never hold for any matrix B . The resolution is precisely Step 1 above: rather than inverting X directly, we first form $X^T X$, which is square ($p \times p$) and invertible, and only then apply the inverse-of-a-product rule legitimately.

Step 2: Invert $X^T X$

Since V is square orthonormal and D^2 is diagonal with positive entries, all three factors in $VD^2 V^T$ are square and invertible. Applying the inverse-of-a-product rule to three factors and using $V^{-1} = V^T$ and $(V^T)^{-1} = V$:

$$(X^T X)^{-1} = (VD^2 V^T)^{-1} = (V^T)^{-1} (D^2)^{-1} V^{-1} = V D^{-2} V^T.$$

Step 3: Compute $(X^T X)^{-1} X^T$

Substituting $X^T = (UDV^T)^T = VD U^T$ and applying the result of Step 2:

$$(X^T X)^{-1} X^T = VD^{-2} V^T \cdot VD U^T = VD^{-2} \underbrace{(V^T V)}_{=I_p} D U^T = VD^{-2} D U^T = VD^{-1} U^T,$$

where $V^T V = I_p$ is square-case orthonormality of V , and $D^{-2} D = D^{-1}$ follows from the diagonal exponent rule.

Step 4: Conclude

$$\boxed{\hat{\beta} = (X^T X)^{-1} X^T Y = VD^{-1} U^T Y.}$$